EXERCISE SHEET 2

GALOIS THEORY

2.1. Let K be a finite field of characteristic p, for a prime p.

(a) Show that $p \mid \binom{p}{i}$ for all $1 \leq i \leq p-1$.

(b) Show that for a finite field K of characteristic p, we have $(a \pm b)^p = a^p \pm b^p$ for any $a, b \in K$.

(c) Show that for a finite field K of characteristic p the mapping $x \mapsto x^p$ is an isomorphism.

2.2. Use the Eisenstein's criterion to prove that the following polynomials are irreducible over $\mathbb{Q}$:

(a) $t^7 + 12t^3 - 22t + 6$;

(b) $t^4 + t^3 + t^2 + t + 1$ (*hint:* consider $g(x) = f(x+1)$);

(c) $t^3 + 77$.

2.3. Prove that an irreducible polynomial over $\mathbb{R}$ has degree ≤ 2. [*Hint:* show that the map $a + ib \mapsto a - ib$ is an automorphism of $\mathbb{C}$ and hence $f(a+ib) = 0 \Rightarrow f(a-ib) = 0$ for $f(t) \in \mathbb{R}[t]$.]

2.4. Find a splitting field $K : \mathbb{Q}$ for each of the following polynomials, and calculate $[K : \mathbb{Q}]$ in each case:

(a) $x^4 - 5x^2 + 6$; (b) $x^4 - 7$; (c) $x^3 - 2$; (d) $x^4 + 4$.

2.5. Let $L : K$ be a finite and normal extension. By the theorem on finite extensions, $L = K(a_1, \ldots, a_s)$ for some a_i algebraic over K. Let $m_i \in K[t]$ be the minimal polynomial of a_i and put $f = m_1 \cdots m_s$. Show that L is a splitting field for f.

Extra questions

2.6. Let $K(t)$ be a simple transcendental extension of a field K. Prove that $K(t^2+1)$ is also a simple transcendental extension of K.